

Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We use a fixed one-minute calendar-time grid from 09:30 to 15:59 for each trading day. This corresponds to 390 one-minute observations per normal trading day. Since a stock may not trade in every minute, we handle missing prices using previous-tick interpolation within each trading day only. This means that if AAPL has no observed price in a given minute, we carry forward the most recently observed price from the same trading day. We do not carry prices across trading days, and we do not backfill prices before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, we sample prices on a fixed grid relative to the market open. For example, with 10-minute sampling, prices are taken at 09:30, 09:40, 09:50, and so on. This differs from a bucket-based approach, where the last observed price within each Δ -minute interval is used. We prefer the fixed-grid approach because it makes the sampling rule explicit and because it will later allow us to align intraday returns across all 30 stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where $r_{d,j}^{(\Delta)}$ is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

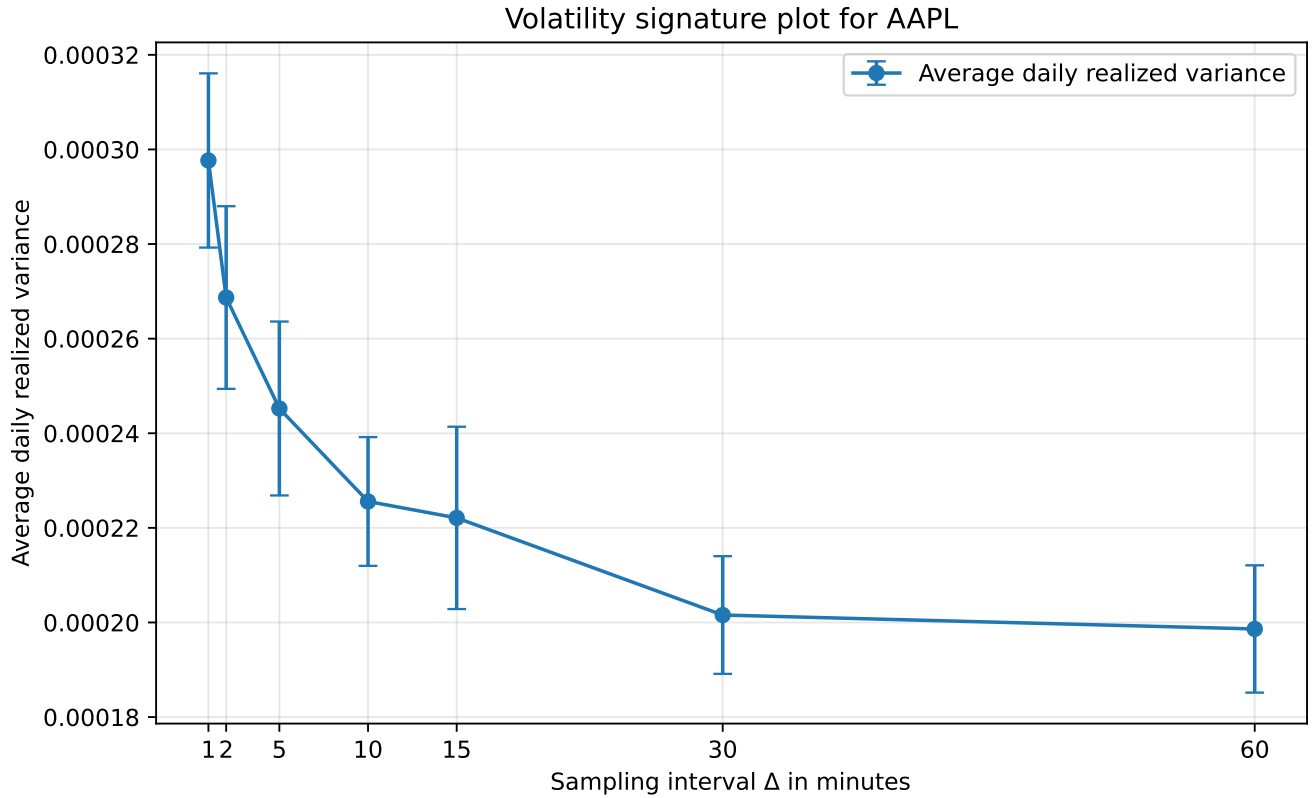


Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

Exercise 2a: Daily realized covariance matrices

We extend the realized variance estimator from Exercise 1 to the multivariate setting and estimate a daily realized covariance matrix for the 30 DJIA constituents.

Let $r_{d,j}$ denote the 30×1 vector of synchronized 5-minute log returns at intraday interval j on trading day d . The realized covariance matrix is computed as

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j}.$$

This is the multivariate analogue of the realized variance estimator used in Exercise 1.

The data are available at the minute frequency rather than at the transaction level. To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day. Missing prices are handled using previous-tick interpolation (forward filling), meaning that the most recently observed price within the same trading day is carried forward. Prices are never carried across trading days.

After synchronization, prices are sampled every five minutes and converted into log returns. The 5-minute frequency is chosen based on the volatility signature plot in Exercise 1 and is also consistent with the realized-volatility literature, where 5-minute returns are commonly used as a compromise between reducing market microstructure noise and retaining intraday information.

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i , the diagonal element is

$$RC_{d,ii} = \sum_{j=1}^{M_d} r_{d,j,i}^2.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

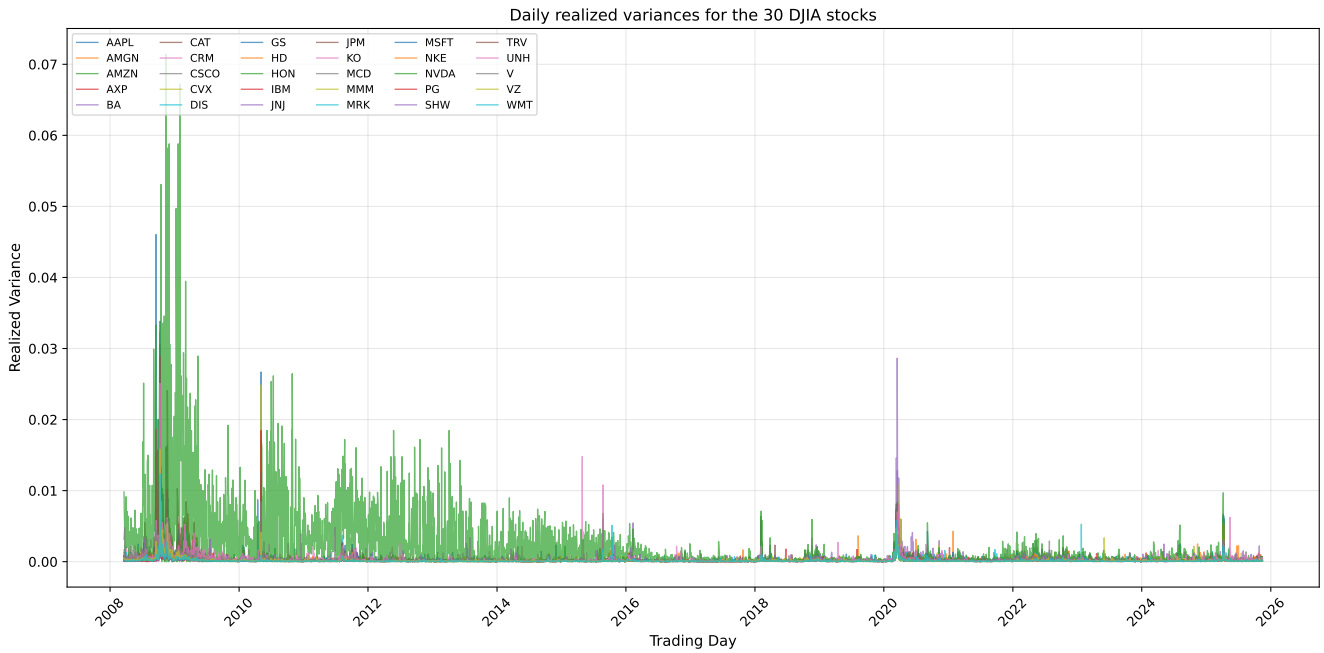


Figure 2: Daily realized variances for the 30 DJIA stocks.

Exercise 3: Covariance estimators

For each day t in the second half of the sample, we construct two estimators of today's covariance matrix using only information available up to day $t - 1$.

Exercise 3a-b: High-frequency random walk and Ledoit-Wolf shrinkage estimator

The high-frequency estimator is defined as:

$$\hat{\Sigma}_t^{HF} = RC_{t-1}.$$

Thus, yesterday's realized covariance matrix is used as today's covariance forecast. This implies a random-walk assumption for the covariance matrix: the best forecast of today's covariance matrix is yesterday's realized covariance matrix. This assumption is motivated by the persistence and clustering typically observed in volatility and covariance dynamics.

Exercise 3c: Comparing realized and Ledoit-Wolf variances

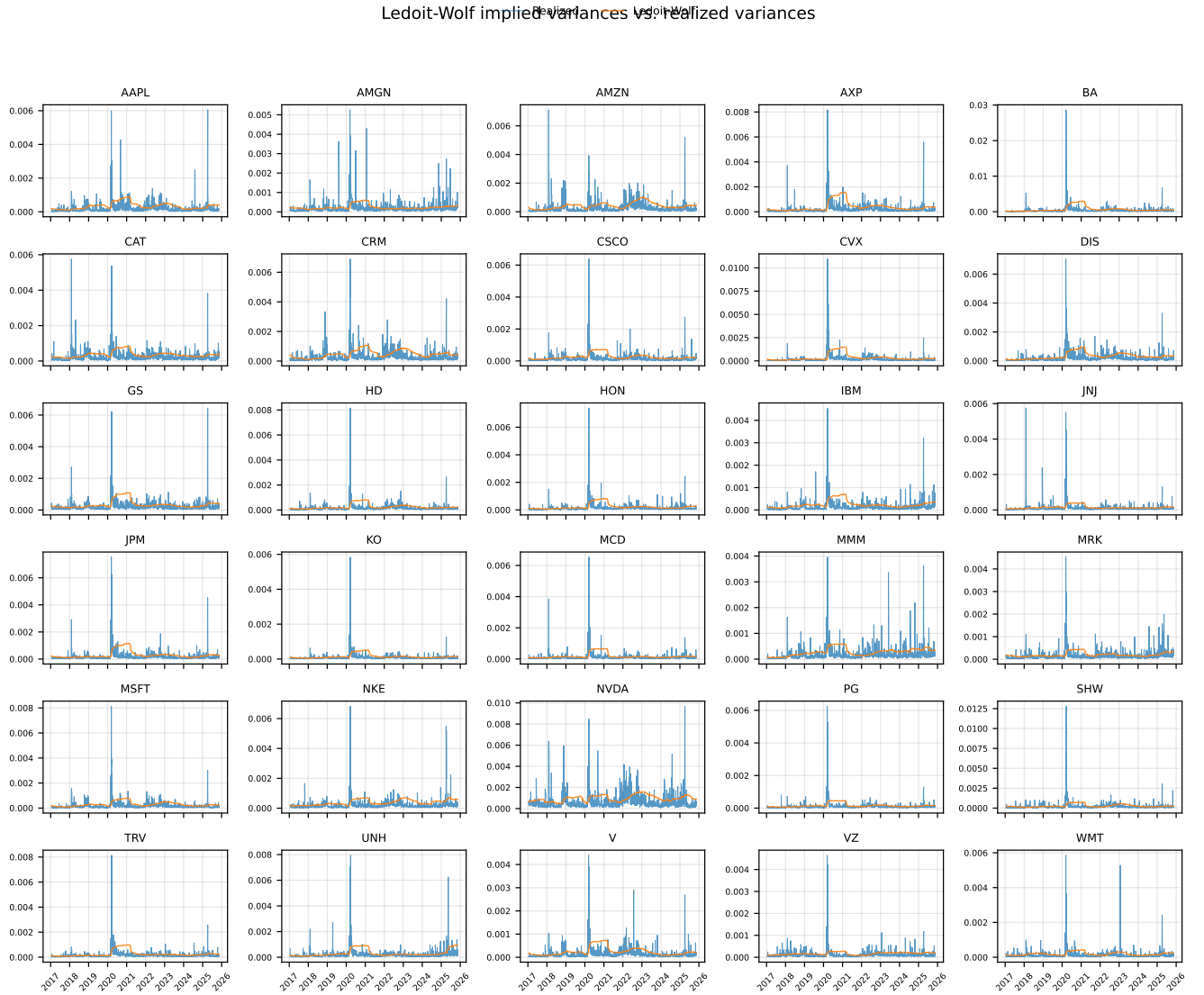


Figure 3: Ledoit-Wolf implied variances and realized variances for the 30 DJIA stocks. The Ledoit-Wolf estimates are based on a 252-day rolling window of daily close-to-close returns, while realized variances are the diagonal elements of the daily realized covariance matrices.

Exercise 4

For each day (t) in the evaluation window, we compute unconstrained minimum-variance portfolio weights for the two covariance estimators from Exercise 3. The weights are computed as

$$\omega_t = \frac{\hat{\Sigma}_t^{-1} \mathbf{1}}{\mathbf{1}' \hat{\Sigma}_t^{-1} \mathbf{1}},$$

where $\mathbf{1}$ is a vector of ones. We compute one set of weights using the high-frequency covariance estimator, $\hat{\Sigma}_t^{HF}$, and one set using the Ledoit-Wolf estimator, $\hat{\Sigma}_t^{LW}$. The weights are fully invested, but short positions are allowed.

Table 1: Turnover between rebalances for the high-frequency and Ledoit-Wolf minimum-variance strategies.

Table 1

Strategy	Average turnover (%)	Median turnover (%)	Max turnover (%)	Rebalances
HF	349.650000	328.440000	934.560000	2222
LW	7.120000	4.990000	98.450000	2222

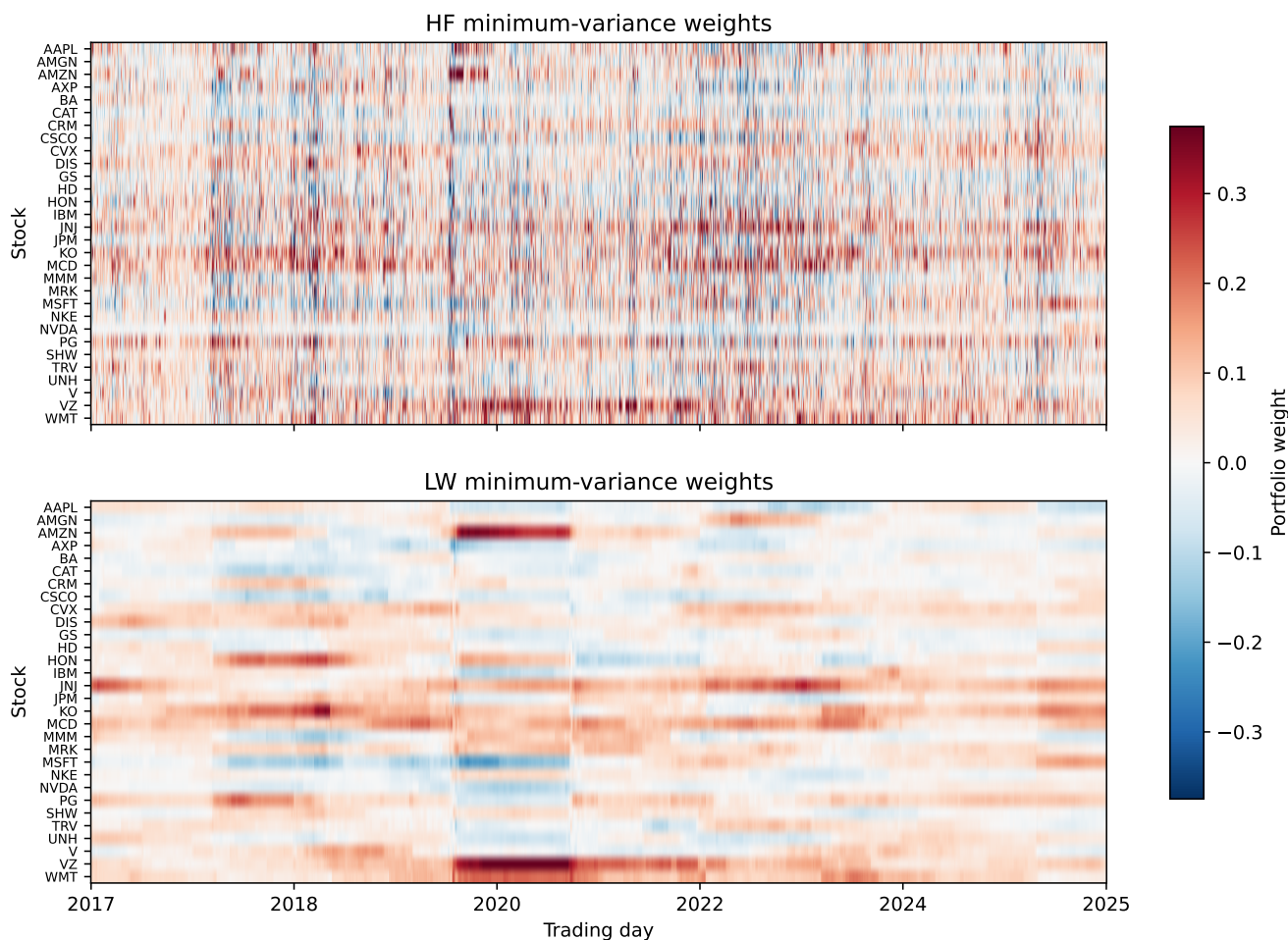


Figure 4: Minimum-variance portfolio weights for the high-frequency (top) and Ledoit-Wolf (bottom) covariance estimators. Rows are stocks, columns are rebalancing dates. Red = long, blue = short, white = zero. Both panels share the same colour scale, capped at the 99th percentile of absolute weights so that typical variation is visible; a few extreme cells saturate.

Exercise 5

We evaluate the three portfolio strategies out of sample over the same evaluation window as in Exercise 3. For each day t , we apply the portfolio weights estimated using only information available up to day $t - 1$ to the realized close-to-close return on day t . This gives realized out-of-sample portfolio returns for the high-frequency minimum-variance strategy, the Ledoit-Wolf minimum-variance strategy, and an equal-weighted benchmark.

For a given strategy with weights ω_t , the realized portfolio return is

$$r_{p,t} = \omega_t' r_t,$$

where r_t is the vector of realized daily close-to-close returns on day t . Since a portfolio return is a weighted average of simple returns, we convert the daily log returns to simple returns before

applying the weights. We report annualized portfolio volatility and annualized Sharpe ratios, assuming a zero risk-free rate.

Table 2: Out-of-sample performance of the high-frequency, Ledoit-Wolf, and equal-weighted strategies.

Strategy	Annualized return (%)	Annualized volatility (%)	Annualized Sharpe ratio
Equal-weighted	17.7503	17.8723	0.9932
HF	11.8208	16.5841	0.7128
LW	11.3662	13.9929	0.8123

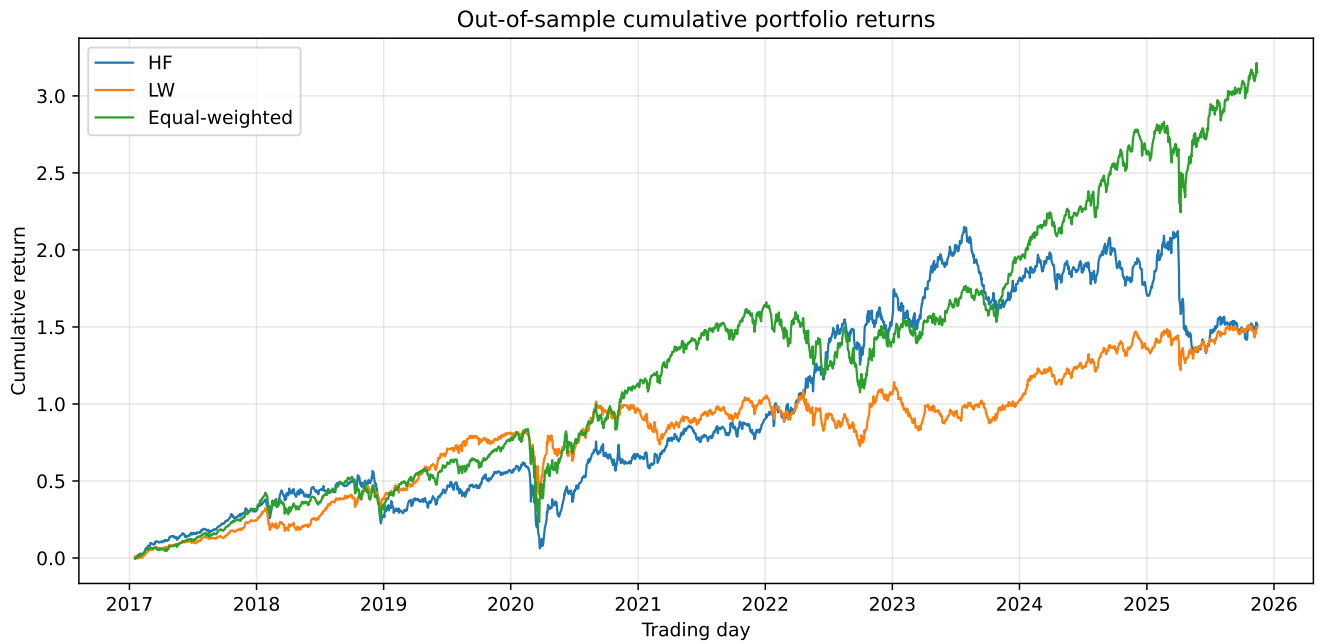


Figure 5: Cumulative out-of-sample portfolio returns for the high-frequency, Ledoit-Wolf, and equal-weighted strategies.

Table 2 reports the out-of-sample results. The Ledoit-Wolf strategy attains the lowest annualized volatility at 13.99% with a Sharpe ratio of 0.812, whereas the high-frequency strategy is both more volatile, at 16.58%, and lower-Sharpe, at 0.713. The equal-weighted benchmark has the highest volatility, 17.87%, but also the highest Sharpe ratio at 0.993.

The high-frequency estimator therefore does not deliver a measurable advantage over Ledoit-Wolf shrinkage (Ledoit and Wolf, 2003, 2004). On the contrary, it is dominated on both dimensions: it is more volatile and has a lower Sharpe ratio than the shrinkage estimator, even before trading costs are taken into account. This is notable given that both are minimum-variance portfolios: Ledoit-Wolf achieves the lowest realized volatility of all three strategies, exactly as the minimum-variance objective intends, while the high-frequency estimator does not. The difference is the same instability documented in Exercise 4, where the high-frequency strategy rebalanced with an average turnover of roughly 349% against only 7% for Ledoit-Wolf. Once realistic transaction costs are

accounted for, the small amount of risk the high-frequency estimator does remove would be far outweighed by the cost of its trading, so the richer intraday information it exploits is not translated into an economically meaningful improvement. As seen in Figure 5, the high-frequency strategy also displays the most violent swings in cumulative return, including a sharp drawdown in 2025, while Ledoit-Wolf is the smoothest of the three.

The broader result is that neither optimized strategy beats the naive equal-weighted benchmark on a risk-adjusted basis: the benchmark, which estimates nothing and therefore carries no estimation error, attains the highest Sharpe ratio of the three. Whether an estimator that exploits additional information — intraday returns, shrinkage targets, or factor structures — improves on equal weighting depends on whether its reduction in estimation error outweighs the noise it introduces through the inversion of the covariance matrix. As emphasised in the in-class discussion and in the shrinkage literature (?), the sample covariance matrix is poorly conditioned when the number of assets is large relative to the sample, and imposing structure can substantially improve out-of-sample behaviour; this is precisely why Ledoit-Wolf outperforms the high-frequency estimator here. We would expect sophistication to pay off when the sample is long relative to the number of assets, when the covariance structure is stable and persistent so that past estimates forecast future risk well, and when turnover is controlled or trading costs are low. The minimum-variance problem with thirty assets and a covariance matrix re-estimated daily sits at the opposite end of this spectrum: estimation error dominates, structure (Ledoit-Wolf) helps more than extra information (high-frequency data), and even then the simplest possible rule is hard to beat. Our evidence suggests that statistical sophistication is valuable in portfolio choice mainly insofar as it stabilises the inputs to optimisation, and that sophistication which merely increases responsiveness — as with the daily-updated high-frequency estimator — does not reliably beat a simple benchmark once realistic frictions are considered.

References

- Ledoit, O. and Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5):603–621.
- Ledoit, O. and Wolf, M. (2004). Honey, i shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119.